

gemstone-rs Release Checklist

Crates, VSIX, GitHub release, and verification steps



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Release Checklist

Use this checklist for coordinated crate, VSIX, and GitHub releases.

0.2.1 / Workbench 0.3.1 Notes

- CLI setup: global `--env-file`, `doctor --strict`, and safer `.env.gemstone-rs` template loading across CLI and explorer startup.
- Codegen: `codegen explain --json`, schema files, and a generated-wrapper compile smoke test.
- Mapping: missing keys, nested dictionaries, and array reads now report richer path context such as `booking.items[2]`.
- Explorer: setup assistant, env sample/write, and a structured Codegen Explain action.
- VS Code: `GemStone RS: Verify Strict Setup` and automatic env-file passing when `gemstoneRs.envFile` exists.

Workbench 0.3.2 Notes

- VS Code: `GemStone RS: Run Setup Assistant` renders the explorer `/api/setup/assistant` checks in the output panel.
- CI: schema copies and `codegen explain --json` output are validated by `scripts/validate_codegen_schemas.js`.
- CI: `compare ... --json`, including status, scorecard, and parity views, is validated against the same structural contract documented in `schemas/gemstone-rs.compare.schema.json`.
- Examples: the generated-wrapper smoke crate now runs `cargo test`.

Workbench 0.3.3 Notes

- CLI: `codegen explain-profile [--json] <profile-name> [profile-file]`
resolves project profiles before rendering codegen summaries.
- VS Code: `GemStone RS: Codegen Explain` and ``GemStone RS: Codegen Explain Profile`` render structured classes, selectors, mappings, and test stubs.

0.2.2 / Workbench 0.3.4 Notes

- Core: project profile check reports now live in `gemstone-rs` and are shared by CLI, explorer, and VS Code.
- Explorer: `/api/codegen/profiles/check` returns the shared JSON report and the browser renders a profile status table with Preview, Diff, Check, and Generate actions.
- Tooling: `schemas/gemstone-rs.profile-check.schema.json` documents the shared report shape, and CI runs a real explorer HTTP endpoint smoke test.
- Tooling: `scripts/version_check.py` now covers release command snippets, workflow defaults, VSIX filename examples, and generated Axum/Actix scaffold adapter versions.

Before Release

- Update crate versions in `Cargo.toml` files.
- Update `vscode-gemstone-rs-workbench/package.json`.
- Regenerate codegen examples:

```
cargo run -p gemstone-rs-cli -- codegen generate examples/codegen/gemstone-rs.codegen
cargo run -p gemstone-rs-cli -- codegen check-profile default examples/codegen/
gemstone-rs.codegen-profiles.json
cargo run -p gemstone-rs-cli -- profile validate examples/codegen/gemstone-
rs.codegen-profiles.json
cargo run -p gemstone-rs-cli -- profile validate --json examples/codegen/gemstone-
rs.codegen-profiles.json
cargo run -p gemstone-rs-cli -- profile list --json examples/codegen/gemstone-
rs.codegen-profiles.json
cargo run -p gemstone-rs-cli -- profile show default --json examples/codegen/
gemstone-rs.codegen-profiles.json
cargo run -p gemstone-rs-cli -- profile resolve default --json examples/codegen/
gemstone-rs.codegen-profiles.json
```

```
cargo run -p gemstone-rs-cli -- profile check examples/codegen/gemstone-rs.codegen-
profiles.json
cargo run -p gemstone-rs-cli -- profile check --json examples/codegen/gemstone-
rs.codegen-profiles.json
cargo run -p gemstone-rs-cli -- codegen explain-profile --json default examples/
codegen/gemstone-rs.codegen-profiles.json
cargo run -p gemstone-rs-cli -- profile sample > /tmp/gemstone-rs.codegen-
profiles.json
cargo test --manifest-path examples/codegen-wrapper-check/Cargo.toml
diff -u examples/codegen/gemstone-rs.codegen-profiles.json /tmp/gemstone-rs.codegen-
profiles.json
```

- Refresh screenshots when the explorer or workbench UI changed:

```
make screenshots
make visual-asset-check
make docs-link-check
make docs-index-check
```

- Run local verification:

```
python3 scripts/version_check.py
python3 scripts/crate_metadata_check.py
make verify
make vscode-package
python3 docs/build_pdf_docs.py
```

Or use the dry-run release wrapper:

```
DRY_RUN=1 scripts/release_all.sh 0.2.2
```

`make verify` includes version and crate metadata checks. The version check guards release workflow defaults, command snippets, VSIX filename examples, and generated Axum/Actix scaffold adapter versions. The crate metadata check also confirms the root README layout lists every publishable crate. The same verification pass then checks that PDF generation completes, produces non-empty PDF files, and exactly matches the configured PDF target set. The release wrapper writes repository-relative SHA256 entries and verifies the expected VSIX plus every PDF with `scripts/verify_release_artifacts.py`. `make verify` also runs an offline release-asset verifier smoke test so checksum mismatches and missing downloaded assets stay covered without contacting GitHub. The release workflow rebuilds and attaches fresh PDFs for the target runner because WeasyPrint output can differ byte-for-byte across platforms. The VSIX filename uses `vscode-gemstone-rs-workbench/package.json`; the crate release tag still uses the workflow `version` input.

Commit Review Pass

Before publishing, make a deliberate release commit review pass:

```
git status --short
git diff --stat
make verify
```

Confirm the release commit includes any regenerated PDFs, refreshed screenshots, profile schema changes, sample profile updates, and VSIX docs. Keep publishing for a separate step unless the release workflow is intentionally being run from that commit.

Publish

Set GitHub secrets once:

```
gh secret set CARGO_REGISTRY_TOKEN
gh secret set VSCE_PAT
```

Run the release workflow:

```
gh workflow run release.yml \
  --ref main \
  -f version=0.2.2 \
  -f publish-crates=true \
  -f publish-vsix=true \
  -f create-github-release=true \
  -f dry-run=false
```

The workflow publishes crates in dependency order:

1. gemstone-gci
2. gemstone-rs-macros
3. gemstone-rs
4. gemstone-rs-axum
5. gemstone-rs-actix
6. gemstone-rs-cli
7. gemstone-rs-explorer

It also packages/publishes the VSIX and can create the GitHub release with the VSIX attached.

Manual crate publishing remains available:

```
scripts/publish_crates.sh
```

For a dry run:

```
DRY_RUN=1 scripts/publish_crates.sh
```

Manual end-to-end publishing remains available through the release wrapper:

```
PUBLISH_CRATES=1 PUBLISH_VSIX=1 CREATE_GITHUB_RELEASE=1 VERIFY_PUBLIC=1 DRY_RUN=0  
scripts/release_all.sh 0.2.2
```

The GitHub **Release** workflow now calls the same `scripts/release_all.sh` wrapper, so local and Actions releases share the same verification, packaging, checksum, publish, GitHub release, and post-publish verification path.

Dry-run mode checks each crate's publish file list locally. The real publish path still uses `cargo publish` in dependency order so crates.io resolves each new dependency after it has been published. If a workflow is rerun after a partial publish, already-published crate versions are skipped and the remaining crates continue.

Post-Release Verification

```
scripts/publish_verify.sh 0.2.2
```

The script checks:

- crates.io package/version JSON for all crates
- `cargo install gemstone-rs-cli`
- `cargo install gemstone-rs-explorer`
- `gemstone-rs --help`
- `gemstone-rs-explorer --help`
- VS Code Marketplace version matches `package.json`
- GitHub Release `v<version>` exists
- GitHub Release assets include the VSIX, `SHA256SUMS`, and every generated PDF
- downloaded GitHub Release assets match the published `SHA256SUMS`

To skip the GitHub Release asset check during early testing:

```
VERIFY_GITHUB_RELEASE=0 scripts/publish_verify.sh 0.2.2
```

The scheduled/manual **Post-Release Verify** workflow runs the same public verification against crates.io, the VS Code Marketplace, and GitHub Release assets. Run it manually when you want to verify a specific version:

```
gh workflow run post-release-verify.yml \
  --ref main \
  -f version=0.2.2 \
  -f verify-github-release=true \
  -f run-live-smoke=false
```

Optional Live Smoke

Run the manual GitHub Actions live job with GemStone secrets configured, enable `run-live-smoke=true` on the post-release workflow, or run locally:

```
scripts/live_smoke.sh --dry-run
scripts/live_smoke.sh
```

The live smoke coverage includes `doctor --strict --live`, login/logout, `3 + 4 == 7`, global put/get, string round-trip, `perform`, commit, abort, browser lookup of `Object`, generated wrapper `printString`, the `py-native smoke` shared-core check, the `live_smoke_cookbook` example, the `python_native_adapter` shared-core example, and live Axum/Actix `/health/gemstone` routes.

The script runs the Rust live tests with `--test-threads=1`. GemStone GCI has process-global session state, so live tests should not be run concurrently.

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